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LEARNER GUIDE

For

R Fundamental and Statistical Analysis for Beginners

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
14.0	Legacy release	Legacy master trainer slides and original R exercises.	Tertiary Infotech Academy Pte Ltd
15.0	20 August 2026	Complete source-aligned rebuild: visual deck, detailed Learner Guide, slide-mapped Lesson Plan, twelve R labs with Excel workbooks, and aligned WA/Case Study assessments.	Tertiary Infotech Academy Pte Ltd

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How to Use This Guide

Use the slides for concepts and decision frameworks. Use this Learner Guide and each lab folder for detailed procedures, exact R code, evidence requirements and acceptance checks. Work inside one lab folder at a time so relative paths remain valid.

Required tools: R 4.x, RStudio Desktop, Microsoft Excel or LibreOffice, and the readxl package. Courseware and assessments are available through the LMS.

Course Learning Outcomes

- LO1: Install R and RStudio IDE and work confidently with scripts, the console and help system.
- LO2: Select, create, inspect and manipulate R data types and data structures.
- LO3: Use R packages and datasets and import, validate and export external data.
- LO4: Create and interpret appropriate data visualisations in R.
- LO5: Write R programs using conditions, loops and reusable functions.
- LO6: Apply descriptive statistics, correlation, regression, hypothesis testing and ANOVA in R.

Topic 1: Getting Started in R

Coverage: R · RStudio · scripts · comments · help. Mapping: K4 · A2.

What is R?

An open-source language and environment for statistical computing, graphics and reproducible analysis.

How it works	Decision rule	Common pitfall
Objects live in memory; functions transform them; scripts preserve the exact sequence of work.	Use R when the task combines data preparation, statistical reasoning, visualisation and repeatable reporting.	R is case-sensitive: mean(x) and Mean(x) are different names.

R pattern:

```
x <- c(12, 15, 18)
mean(x)
```

Interpretation: A script turns an interactive calculation into an auditable analysis.

Install R and RStudio

R is the computation engine; RStudio is an integrated development environment that makes the engine easier to use.

How it works	Decision rule	Common pitfall
Install R first, then RStudio Desktop. RStudio detects the installed R version.	Confirm both versions and keep packages in a user library that you can update.	Installing only RStudio does not install the R engine.

R pattern:

```
R.version.string
R.home()
```

Interpretation: Verify the engine before troubleshooting packages or scripts.

RStudio Interface

The Source, Console, Environment/History and Files/Plots/Packages/Help panes separate authoring, execution, objects and output.

How it works	Decision rule	Common pitfall
Write durable code in Source; execute selected lines; inspect objects and plots without mixing them into the script.	Save the project and script before running a new analysis.	Commands typed only in Console are easy to lose and difficult to audit.

R pattern:

```
getwd()
ls()
objects()
```

Interpretation: Use the Source pane as the record of analysis; use the Console as feedback.

R Scripts and Comments

An .R script is a plain-text program containing valid R expressions and comments.

How it works	Decision rule	Common pitfall
Run one line, a selection or the whole script. Use # to explain intent and assumptions.	Organise scripts into setup, import, validation, analysis, visualisation and export sections.	Comments should explain why a choice was made, not merely repeat the code.

R pattern:

```
# Average valid measurements
avg <- mean(x, na.rm = TRUE)
```

Interpretation: Readable scripts are easier to review, rerun and repair.

Help and Self-Diagnosis

R includes searchable documentation, examples and object inspection tools.

How it works	Decision rule	Common pitfall
Use help(), ?, args(), example(), str() and traceback() to move from symptom to evidence.	Inspect the function signature and the object structure before changing code.	Copying an online snippet without checking the installed package version can create silent mismatches.

R pattern:

```
help("plot")
args(mean)
str(mtcars)
```

Interpretation: Good R users do not memorise everything; they diagnose efficiently.

Lab 1: Your First Reproducible R Script

Goal: A saved .R script that creates, summarises and exports attendance data.

Scenario: A training coordinator needs a repeatable summary of course attendance.

Mapping: LO1 · K4 · A2 · Level: Foundation

Step-by-step

1. Open the workbook and review the Brief and Input_Data sheets.

2. Create lab-01.R in RStudio and add a header comment with purpose and source.

3. Create the attendance vector and calculate valid count, mean and minimum.

4. Use help('mean') and record the na.rm decision.

5. Export the summary and run verify.R.

Open the folder labs/lab-01-rstudio-first-script/ and run lab-01.R from that working directory.

Acceptance criteria

- Script runs from a clean R session without manual Console steps.
- Summary excludes the missing value and reports the expected valid count.
- Evidence sheet records the output filename and verification result.

Evidence to retain: completed Excel Evidence sheet, generated output files, R console verification message and the final R script.

Lab 2: Help, Objects and Session Diagnostics

Goal: A diagnostic R script and evidence log explaining the root cause.

Scenario: A colleague's script fails because an object and function call are misunderstood.

Mapping: LO1 · K4 · A2 · Level: Foundation

Step-by-step

1. Inspect the workbook's Issue_Log sheet.

2. Use help(), args(), str() and class() on the named objects.

3. Capture sessionInfo() and the active working directory.

4. Correct the failing expression and rerun it.

5. Record diagnosis and evidence in the workbook.

Open the folder labs/lab-02-help-session-diagnostics/ and run lab-02.R from that working directory.

Acceptance criteria

- Diagnostics identify the missing argument or wrong object type.
- Corrected expression returns the expected value.
- Session information is captured for reproducibility.

Evidence to retain: completed Excel Evidence sheet, generated output files, R console verification message and the final R script.

Topic 2: Data Types

Coverage: atomic values · vectors · matrices · arrays · data frames · lists · factors. Mapping: K3 · K6 · K7 · A5 · A9.

Atomic Values and Variables

R stores logical, integer, double, complex, character and raw atomic values.

How it works	Decision rule	Common pitfall
Assignment binds a name to a value; typeof(), class() and str() expose different aspects of the object.	Use descriptive snake_case names and explicit conversion at system boundaries.	Coercion can change the entire vector when one incompatible value is added.

R pattern:

```
count <- 12L
rate <- 0.35
label <- "May"
typeof(rate)
```

Interpretation: Type is a property of the object, not the variable name.

Character Data

Character vectors hold text; quotes delimit values and escaping represents special characters.

How it works	Decision rule	Common pitfall
paste(), paste0(), nchar(), substr(), strsplit() and gsub() create, inspect and transform strings.	Use factors for repeated categorical levels; keep free text as character.	A missing value NA is not the same as the literal string "NA".

R pattern:

```
email <- "learner@example.com"
parts <- strsplit(email, "@")[[1]]
paste(parts, collapse = " at ")
```

Interpretation: Text cleaning should preserve meaning while making categories consistent.

Vectors

A vector is a one-dimensional, homogeneous sequence and the foundation of R computation.

How it works	Decision rule	Common pitfall
c(), :, seq() and rep() create vectors; vectorised operators act element by element.	Prefer vectorised operations for clarity and performance when every element follows the same rule.	Recycling a shorter vector may produce a result that runs but is conceptually wrong.

R pattern:

```
scores <- c(72, 84, 91, 67)
scaled <- scores / 100
mean(scaled)
```

Interpretation: Think in complete columns rather than one cell at a time.

Indexing and Missing Values

Square brackets select by position, name or logical condition; negative indices exclude.

How it works	Decision rule	Common pitfall
is.na() identifies missing values and na.rm = TRUE tells many summaries to omit them.	Make the missing-data rule explicit before calculating a statistic.	Comparing x == NA never identifies missing values; use is.na(x).

R pattern:

```
x <- c(4, NA, 9, -2, 7)
x[!is.na(x) & x > 0]
mean(x, na.rm = TRUE)
```

Interpretation: Filtering is part of the analysis logic and must be documented.

Vector Statistics and Arithmetic

Summary functions collapse a vector; cumulative functions preserve a running path.

How it works	Decision rule	Common pitfall
mean, median, sum, min, max, quantile, sort, diff, cumsum and rank answer different questions.	Choose robust summaries such as median and IQR when outliers distort the mean.	A plausible number can still answer the wrong statistical question.

R pattern:

```
x <- c(10, 11, 12, 13, 80)
c(mean = mean(x), median = median(x), iqr = IQR(x))
```

Interpretation: Compare location and spread before making a claim about typical values.

Matrices

A matrix is a two-dimensional homogeneous object with rows and columns.

How it works	Decision rule	Common pitfall
matrix(), rbind() and cbind() construct matrices; [row, column] selects elements.	Use matrices for numeric linear algebra; use data frames for mixed-type records.	R fills a matrix by column unless byrow = TRUE is specified.

R pattern:

```
m <- matrix(1:12, nrow = 3, byrow = TRUE)
m[2, ]
m[, 3]
```

Interpretation: Always state the dimension and fill direction when constructing a matrix.

Arrays

An array generalises a matrix to three or more dimensions while keeping one atomic type.

How it works	Decision rule	Common pitfall
dim specifies the length of each axis; commas separate the indices for each dimension.	Use an array for regularly shaped repeated measurements such as site × metric × month.	Without dimension names, slices are difficult to interpret and easy to transpose.

R pattern:

```
a <- array(1:18, dim = c(3, 2, 3))
a[, , 2]
```

Interpretation: Name dimensions whenever an axis carries business meaning.

Data Frames

A data frame is a two-dimensional table whose columns may have different types but equal length.

How it works	Decision rule	Common pitfall
$`, [[ ]] and [rows, columns] access data; head(), tail(), names(), str() and summary() reveal structure.	Use a data frame for observational records such as one row per person or day.	Row names are labels, not a reliable database key.

R pattern:

```
cars <- mtcars
small <- cars[c("mpg", "am", "wt")]
head(small)
```

Interpretation: Inspect column names and types before filtering or modelling.

Filtering and Subsetting

Logical expressions select rows that meet one or more conditions.

How it works	Decision rule	Common pitfall
Use &, | and ! element-wise; subset() can improve readability for simple interactive tasks.	Store complex conditions in named variables so the rule can be reviewed.	&& and || examine only the first element and are normally wrong for row filtering.

R pattern:

```
keep <- mtcars$mpg > 15 & mtcars$am == 1
result <- mtcars[keep, c("mpg", "am", "wt")]
summary(result)
```

Interpretation: A filter is a business rule expressed as code.

Lists

A list can hold objects of different types, sizes and shapes.

How it works	Decision rule	Common pitfall
list() creates; [[ ]] extracts one element; [ ] returns a sub-list; $ accesses a named element.	Use lists for model results, nested configuration and heterogeneous outputs.	Confusing [ ] with [[ ]] often leaves a list where a vector or model was expected.

R pattern:

```
result <- list(data = mtcars, model = lm(mpg ~ wt, mtcars))
coef(result$model)
```

Interpretation: Lists let one analysis result carry data, model and diagnostics together.

Factors

A factor represents categorical data as a fixed set of levels.

How it works	Decision rule	Common pitfall
factor() defines levels and order; table() counts; relevel() changes the reference category.	Use factors when a category will drive grouping, plotting or a statistical model.	Unused or misspelled levels can change plots and model coefficients.

R pattern:

```
transmission <- factor(mtcars$am, levels=c(0,1), labels=c("Automatic","Manual"))
table(transmission)
```

Interpretation: Factor levels encode meaning, not just display order.

Choosing the Right Structure

Shape, type consistency and access pattern determine the correct R structure.

How it works	Decision rule	Common pitfall
Vector: 1D homogeneous; matrix/array: regular homogeneous dimensions; data frame: tabular mixed types; list: heterogeneous nesting.	Select the simplest structure that faithfully represents the data and intended operation.	Forcing mixed data into a matrix coerces every value to a common type.

R pattern:

```
objects <- list(vector=1:3, matrix=matrix(1:4,2), frame=mtcars[1:2,1:3])
lapply(objects, class)
```

Interpretation: Data structure is an analytical design decision.

Lab 3: Vectors, Indexing and Missing Values

Goal: A clean vector, filtered subset and robust summary table.

Scenario: An operations analyst must clean sensor readings before reporting an average.

Mapping: LO2 · K3 · A5 · Level: Foundation

Step-by-step

1. Read Input_Data from the workbook using readxl.

2. Create a numeric vector and inspect its type and length.

3. Remove missing values with is.na() and flag readings outside the valid range.

4. Calculate mean, median and IQR.

5. Export the cleaned data and run verification.

Open the folder labs/lab-03-vectors-indexing-missing/ and run lab-03.R from that working directory.

Acceptance criteria

- No invalid or missing reading enters the final mean.
- Mean, median and IQR match the expected values.
- All filtering rules are visible in the R script.

Evidence to retain: completed Excel Evidence sheet, generated output files, R console verification message and the final R script.

Lab 4: Data Structures and Factors

Goal: A typed data frame, a matrix slice and a factor frequency table.

Scenario: A fleet analyst needs the correct R structure for mixed vehicle records.

Mapping: LO2 · K3 · A5 · A9 · Level: Foundation

Step-by-step

1. Import the vehicle table from Excel.

2. Create a numeric matrix for selected measures and extract the specified slice.

3. Create a data frame with numeric and categorical columns.

4. Convert transmission to an ordered factor and produce a frequency table.

5. Bundle data and summaries in a named list.

Open the folder labs/lab-04-structures-factors/ and run lab-04.R from that working directory.

Acceptance criteria

- Matrix contains only numeric values and has the expected dimensions.
- Data frame preserves column types.
- Factor levels and counts are explicit and correct.

Evidence to retain: completed Excel Evidence sheet, generated output files, R console verification message and the final R script.

Topic 3: R Packages and Data I/O

Coverage: packages · built-in datasets · working directories · CSV/Excel · validation · export. Mapping: K3 · K6 · A2 · A5 · A9.

Packages and Libraries

A package bundles functions, data and documentation; a library is the installed location and attached package set.

How it works	Decision rule	Common pitfall
install.packages() installs once; library() attaches per session; package::function calls explicitly.	Prefer explicit package::function in shared scripts when function names may collide.	Running install.packages() on every execution slows work and can change the environment unexpectedly.

R pattern:

```
if (!requireNamespace("readxl", quietly=TRUE)) install.packages("readxl")
library(readxl)
```

Interpretation: Separate environment setup from analysis execution.

Built-in Datasets

R ships documented datasets such as mtcars, quakes, sleep and chickwts for reproducible examples.

How it works	Decision rule	Common pitfall
data() lists available data; ?mtcars explains variables; str() validates structure.	Read the dataset documentation before interpreting a column or unit.	Familiarity with a dataset name does not replace checking its definitions.

R pattern:

```
data(mtcars)
?mtcars
str(mtcars)
```

Interpretation: Metadata is part of the dataset.

Working Directory and Paths

The working directory is the base location used by relative file paths.

How it works	Decision rule	Common pitfall
getwd() reports it; RStudio Projects make it stable; file.path() builds portable paths.	Prefer project-relative paths and avoid setwd() inside reusable scripts.	Absolute user-specific paths break when another learner runs the script.

R pattern:

```
input_path <- file.path("data", "weather.csv")
file.exists(input_path)
```

Interpretation: Portable paths are essential for reproducible course labs.

Import and Validate Data

Import converts external bytes into an R object; validation confirms that the object matches the analysis contract.

How it works	Decision rule	Common pitfall
Read, inspect names/types/dimensions, check missingness and ranges, then analyse.	Stop early when required columns or valid ranges are absent.	A successful read is not proof that dates, categories or numbers were parsed correctly.

R pattern:

```
weather <- read.csv("weather.csv", stringsAsFactors=FALSE)
stopifnot(all(c("Month","Ozone") %in% names(weather)))
summary(weather)
```

Interpretation: Validation protects every downstream chart and model.

Export and Audit

Export writes analysis results in a shareable format while preserving an audit trail.

How it works	Decision rule	Common pitfall
write.csv() writes tables; saveRDS() preserves one R object; save() preserves multiple objects.	Export only the fields needed by the recipient and record the row count and timestamp.	Writing row names to CSV by default creates an unwanted first column.

R pattern:

```
write.csv(result, "analysis_output.csv", row.names=FALSE)
saveRDS(result, "analysis_output.rds")
```

Interpretation: An export should be traceable back to its input and script.

Lab 5: Packages and Built-in Datasets

Goal: A setup script and mtcars/quakes data profile.

Scenario: A junior analyst must document a reproducible package and dataset setup.

Mapping: LO3 · K3 · A2 · Level: Intermediate

Step-by-step

1. Review package requirements in the workbook.

2. Check for readxl without reinstalling it on every run.

3. Load and document mtcars and quakes.

4. Create a compact data profile with rows, columns, missingness and ranges.

5. Write the profile to CSV and log package versions.

Open the folder labs/lab-05-packages-built-in-data/ and run lab-05.R from that working directory.

Acceptance criteria

- Script uses requireNamespace() before optional installation.
- Data profile reports correct dimensions.
- Package and R versions are recorded.

Evidence to retain: completed Excel Evidence sheet, generated output files, R console verification message and the final R script.

Lab 6: Import, Validate and Export Weather Data

Goal: A validated weather dataset and audit-ready summary export.

Scenario: A public-environment team needs a defensible May ozone summary.

Mapping: LO3 · K6 · A2 · A5 · A9 · Level: Intermediate

Step-by-step

1. Open the workbook and note the data contract.

2. Import Input_Data with readxl using a project-relative path.

3. Validate required columns, month values and numeric ozone type.

4. Filter May records and calculate the average ozone with missing values handled.

5. Export analysis_output.csv and update the Evidence sheet.

Open the folder labs/lab-06-import-validate-export/ and run lab-06.R from that working directory.

Acceptance criteria

- Validation stops the script if a required column is absent.
- May average uses only non-missing ozone records.
- Export contains source row count and analysis timestamp.

Evidence to retain: completed Excel Evidence sheet, generated output files, R console verification message and the final R script.

Topic 4: Data Visualisation

Coverage: scatter · line · boxplot · bar · pie · histogram · interpretation. Mapping: K3 · K5 · A3 · A4 · A7 · A10.

Chart Selection

A chart is a visual answer to a specific analytical question, not decoration.

How it works	Decision rule	Common pitfall
Relationship → scatter; trend → line; distribution → histogram/boxplot; category comparison → bar.	Choose the visual encoding that makes the comparison easiest and most accurate.	Starting with a preferred chart type can hide the structure of the data.

R pattern:

```
plot(mpg ~ wt, data=mtcars, pch=19, col="#1F6FEB")
```

Interpretation: State the question first, then choose the chart.

Scatter and Line Plots

Scatter plots reveal association between two numeric variables; lines emphasise ordered progression.

How it works	Decision rule	Common pitfall
plot(x,y) creates points; lines() or type='l' connects an ordered series.	Do not connect unordered observations because the line implies sequence.	Association does not prove causation, even when the point cloud is tight.

R pattern:

```
plot(mtcars$wt, mtcars$mpg, pch=19)
fit <- lm(mpg ~ wt, mtcars)
abline(fit, col="red", lwd=2)
```

Interpretation: Read direction, form, strength and unusual points.

Multiple Scatter Plots

A scatterplot matrix compares every pair among several numeric variables.

How it works	Decision rule	Common pitfall
plot(data_frame) creates the matrix; consistent scales and labels support scanning.	Use it for exploration, then focus the final story on the most decision-relevant pair.	A dense matrix becomes unreadable when too many variables are included.

R pattern:

```
plot(mtcars[c("mpg","disp","hp","wt")])
```

Interpretation: Exploration can be broad; communication should be selective.

Boxplots

A boxplot summarises median, quartiles, spread and potential outliers across groups.

How it works	Decision rule	Common pitfall
The box spans Q1–Q3, the line marks the median and whiskers extend using the 1.5×IQR convention.	Use a boxplot to compare group distributions, not merely their means.	A small group can look deceptively stable; always inspect sample size.

R pattern:

```
boxplot(weight ~ feed, data=chickwts, col="#10B981", las=2)
```

Interpretation: Compare centre, spread, overlap and unusual observations.

Bar Charts

Bar charts compare values or counts across discrete categories from a common zero baseline.

How it works	Decision rule	Common pitfall
table() creates counts; barplot() maps magnitude to bar length.	Sort bars when order is not inherent and keep the baseline at zero.	Using bars for continuous distributions hides shape and variation.

R pattern:

```
counts <- table(mtcars$am)
barplot(counts, names.arg=c("Automatic","Manual"))
```

Interpretation: Bar length is precise; decorative 3D effects are not.

Pie Charts

A pie chart encodes parts of a whole as angles and areas.

How it works	Decision rule	Common pitfall
Counts are converted to proportions that sum to 100%.	Use only for a few categories with a clear whole; prefer bars for close comparisons.	Too many slices and similar sizes make ranking difficult.

R pattern:

```
counts <- table(mtcars$cyl)
pie(counts, main="Cars by cylinders")
```

Interpretation: Use a pie only when the part-to-whole message matters.

Histograms

A histogram groups continuous observations into bins to reveal distribution shape.

How it works	Decision rule	Common pitfall
Breaks control bin boundaries; frequency or density controls the vertical scale.	Try defensible bin widths and report the choice when it affects the story.	A single bin setting can create or hide apparent peaks.

R pattern:

```
hist(mtcars$mpg, breaks=8, col="#7C3AED", border="white")
```

Interpretation: Read centre, spread, skew, modes and unusual gaps.

Lab 7: Base R Chart Selection

Goal: A scatter plot, bar chart and histogram saved as PNG files.

Scenario: A transport team needs three views of fuel-economy and transmission data.

Mapping: LO4 · K3 · A3 · A4 · Level: Intermediate

Step-by-step

1. Import the workbook's vehicle data.

2. State one question for each required chart.

3. Create a labelled scatter plot of weight versus fuel economy.

4. Create a zero-based bar chart of transmission counts and a histogram of fuel economy.

5. Save the plots and write one evidence-based interpretation per chart.

Open the folder labs/lab-07-base-r-chart-selection/ and run lab-07.R from that working directory.

Acceptance criteria

- Each chart answers a distinct question.
- Axes, units and categories are readable.
- Interpretations describe evidence without claiming causation.

Evidence to retain: completed Excel Evidence sheet, generated output files, R console verification message and the final R script.

Lab 8: Compare Distributions with Boxplots

Goal: A grouped boxplot, descriptive table and visual interpretation.

Scenario: An agriculture team compares chick weights across feed groups.

Mapping: LO4 · K5 · A3 · A4 · A7 · A10 · Level: Intermediate

Step-by-step

1. Import the feed dataset from the workbook.

2. Confirm feed is a factor and count observations per group.

3. Create a grouped boxplot with readable labels.

4. Calculate median and IQR for every feed.

5. Identify one unusual group and document the evidence.

Open the folder labs/lab-08-distribution-comparison/ and run lab-08.R from that working directory.

Acceptance criteria

- Boxplot includes every feed group.
- Summary table agrees with the plotted medians and spread.
- Interpretation distinguishes observation from causal claim.

Evidence to retain: completed Excel Evidence sheet, generated output files, R console verification message and the final R script.

Topic 5: R Programming

Coverage: conditions · loops · next/break · operators · functions · arguments. Mapping: K6 · K7 · A7 · A9.

Conditions

if and else choose which block of code runs based on one TRUE/FALSE condition.

How it works	Decision rule	Common pitfall
Evaluate a scalar condition; run the first matching branch; use ifelse() for vectorised element-wise choices.	Make branches mutually understandable and test boundary values.	Using a vector condition in if() checks only length-one logic and may error.

R pattern:

```
risk <- 0.72
label <- if (risk >= 0.7) "High" else "Review"
```

Interpretation: A condition converts a rule into an explicit decision.

While Loops

A while loop repeats as long as its condition remains TRUE.

How it works	Decision rule	Common pitfall
Initialise state, test the condition, perform work, then update state.	Use when the number of repetitions is not known in advance.	If state never changes, the loop can run forever.

R pattern:

```
x <- 1
while (x <= 5) {
  print(x)
  x <- x + 1
}
```

Interpretation: Every while loop needs a clear exit path.

For Loops

A for loop processes each element of a known sequence.

How it works	Decision rule	Common pitfall
The loop variable takes one value at a time; results should normally be preallocated.	Use a loop when each iteration has stateful or multi-step logic that is clearer than vectorisation.	Growing an object repeatedly inside the loop is inefficient.

R pattern:

```
out <- numeric(5)
for (i in seq_along(out)) {
  out[i] <- i^2
}
```

Interpretation: Clarity matters more than avoiding loops at all costs.

next and break

next skips the remainder of the current iteration; break exits the loop completely.

How it works	Decision rule	Common pitfall
Both alter control flow and should sit beside a clearly stated condition.	Use next for invalid records; use break when the goal has been reached or continued work is unsafe.	An early exit without a message can make missing output look like a successful run.

R pattern:

```
for (x in c(3, NA, 7, 12)) {
  if (is.na(x)) next
  if (x > 10) break
  print(x)
}
```

Interpretation: Explain why records are skipped or processing stops.

Operators

Arithmetic, comparison and logical operators form expressions that transform data and test rules.

How it works	Decision rule	Common pitfall
Use + - * / ^ %%, comparisons, and element-wise &, |, !; parentheses make precedence explicit.	Vectorised logical operators are normally required for data filtering.	Floating-point equality can fail due to representation; compare with a tolerance.

R pattern:

```
keep <- !is.na(mtcars$mpg) & mtcars$mpg >= 20
mtcars[keep, ]
```

Interpretation: Readable expressions reduce logical defects.

Functions and Arguments

A function packages reusable behaviour behind named inputs and a return value.

How it works	Decision rule	Common pitfall
function(...) defines; default and named arguments make calls flexible; ... forwards optional arguments.	Use a function when the same transformation appears twice or needs its own test.	A missing required argument causes an error; a bad default can hide an assumption.

R pattern:

```
summarise_scores <- function(x, trim=0) {
  mean(x, trim=trim, na.rm=TRUE)
}
summarise_scores(c(1,2,100), trim=.1)
```

Interpretation: A small, focused function is easier to test than repeated code.

Lab 9: Conditions, Loops, next and break

Goal: A controlled processing loop with an exception log.

Scenario: A quality team scans daily temperatures and handles invalid records.

Mapping: LO5 · K6 · A7 · A9 · Level: Intermediate

Step-by-step

1. Import daily readings from Excel.

2. Use a for loop to count valid days above the threshold.

3. Use next for missing readings and record them.

4. Use break only when the safety-stop value appears.

5. Compare the loop result with a vectorised validation check.

Open the folder labs/lab-09-conditions-loops/ and run lab-09.R from that working directory.

Acceptance criteria

- Missing values are skipped and counted.
- Safety stop is explicit and testable.
- Loop and vectorised count agree before the stop point.

Evidence to retain: completed Excel Evidence sheet, generated output files, R console verification message and the final R script.

Lab 10: Build and Test Reusable Functions

Goal: Two documented functions with deterministic tests.

Scenario: A game-based learning team needs a reliable two-dice simulator and summary function.

Mapping: LO5 · K7 · A7 · A9 · Level: Intermediate

Step-by-step

1. Define roll_dice(n=2, sides=6) with input validation.

2. Use sample() and sum() to return a roll total.

3. Define summarise_rolls(x, ...) and forward options to mean().

4. Set a random seed and run deterministic tests.

5. Record expected and actual results in the workbook.

Open the folder labs/lab-10-reusable-functions/ and run lab-10.R from that working directory.

Acceptance criteria

- Invalid n or sides values stop with clear messages.
- Default and named argument calls both work.
- Tests are reproducible under the recorded seed.

Evidence to retain: completed Excel Evidence sheet, generated output files, R console verification message and the final R script.

Topic 6: Statistical Analysis with R

Coverage: descriptive statistics · correlation · regression · tests · ANOVA · diagnostics. Mapping: K1 · K2 · K5 · A1 · A3 · A4 · A6 · A7 · A8.

Descriptive Statistics

Descriptive statistics summarise observed data without generalising beyond it.

How it works	Decision rule	Common pitfall
Location: mean/median; spread: range/IQR/SD; shape: skew and unusual values; counts: n and missingness.	Report centre, spread and valid sample size together.	Averages without context can conceal missing data or a highly skewed distribution.

R pattern:

```
x <- c(12, 14, 15, 18, NA, 44)
c(n=sum(!is.na(x)), mean=mean(x,na.rm=TRUE), median=median(x,na.rm=TRUE),
sd=sd(x,na.rm=TRUE))
```

Interpretation: Describe the data before modelling it.

Trimmed Mean and Robustness

A trimmed mean removes equal proportions from both tails before averaging.

How it works	Decision rule	Common pitfall
Increasing trim reduces the influence of extreme values while still using more information than the median.	Use a robust summary when outliers are plausible but should not dominate the typical value.	Trimming should be justified and cannot replace investigating data quality.

R pattern:

```
ozone <- airquality$Ozone
c(mean=mean(ozone,na.rm=TRUE), trim10=mean(ozone,trim=.1,na.rm=TRUE),
median=median(ozone,na.rm=TRUE))
```

Interpretation: Robustness is a deliberate trade-off between sensitivity and stability.

Correlation

Correlation measures the strength and direction of a linear relationship between two numeric variables.

How it works	Decision rule	Common pitfall
Pearson r ranges from -1 to +1; sign gives direction and magnitude gives linear strength.	Inspect a scatter plot and outliers before interpreting r.	Correlation does not establish causation and may miss non-linear relationships.

R pattern:

```
r <- cor(mtcars$wt, mtcars$mpg)
round(r, 2)
```

Interpretation: A negative r means one variable tends to decrease as the other increases.

Linear Regression

Simple linear regression estimates a line that predicts an outcome from one input.

How it works	Decision rule	Common pitfall
lm(y ~ x) estimates intercept and slope by minimising squared residuals.	Use when the relationship is plausibly linear and diagnostics support the assumptions.	Extrapolating beyond the observed x range can produce implausible predictions.

R pattern:

```
model <- lm(mpg ~ wt, data=mtcars)
coef(model)
predict(model, data.frame(wt=3))
```

Interpretation: The slope quantifies the expected change in y for one unit of x.

Residuals and Mean Squared Error

A residual is observed minus predicted; MSE averages squared prediction errors.

How it works	Decision rule	Common pitfall
Squaring makes errors positive and penalises large misses more strongly.	Inspect residual plots as well as one error metric.	A low training MSE does not guarantee accurate new predictions.

R pattern:

```
model <- lm(mpg ~ wt, mtcars)
res <- residuals(model)
mean(res^2)
```

Interpretation: Residual patterns reveal model problems that one score can hide.

Regression Diagnostics

Diagnostics test whether the linear model is an adequate representation of the data.

How it works	Decision rule	Common pitfall
Check linearity, constant variance, normality of residuals and influential observations.	Change the model or interpretation when assumptions fail materially.	A significant coefficient is not proof that the full model is useful.

R pattern:

```
model <- lm(mpg ~ wt, mtcars)
par(mfrow=c(2,2))
plot(model)
```

Interpretation: Diagnose unintended outcomes before trusting a model.

Multiple Regression

Multiple regression predicts one outcome from several inputs and can include interactions.

How it works	Decision rule	Common pitfall
Each coefficient is conditional on the other variables; x1:x2 represents an interaction.	Choose predictors from the question and domain logic, not only automated significance.	Highly correlated predictors make coefficients unstable and difficult to interpret.

R pattern:

```
model <- lm(mpg ~ wt + hp + disp + wt:hp, mtcars)
summary(model)
predict(model, data.frame(wt=2.6,hp=110,disp=160))
```

Interpretation: More predictors do not automatically mean a better model.

R-squared and Model Fit

R-squared is the proportion of observed outcome variance explained by the fitted model in the sample.

How it works	Decision rule	Common pitfall
Adjusted R-squared penalises unnecessary predictors; both must be read with residual diagnostics.	Judge usefulness against the decision context, not a universal threshold.	A high R-squared can coexist with bias, leakage or poor out-of-sample performance.

R pattern:

```
model <- lm(mpg ~ wt + hp, mtcars)
summary(model)$r.squared
summary(model)$adj.r.squared
```

Interpretation: Model fit is evidence, not a guarantee of accuracy.

Hypothesis Testing

A hypothesis test evaluates how compatible the observed data are with a stated null hypothesis.

How it works	Decision rule	Common pitfall
Define H0/H1, choose alpha and test, compute a statistic and p-value, then decide and interpret.	Use 'reject' or 'fail to reject' H0; do not claim that a non-significant result proves equality.	Statistical significance is not the same as practical importance.

R pattern:

```
alpha <- 0.05
p_value <- 0.032
if (p_value < alpha) "Reject H0" else "Fail to reject H0"
```

Interpretation: Predefine the question and threshold before looking at the result.

Two-Sample t-Test

A two-sample t-test compares the means of two independent groups.

How it works	Decision rule	Common pitfall
Welch's test is the R default and does not assume equal variances; alternative controls two- or one-sided questions.	Use a two-sided test unless a directional hypothesis was defined in advance.	Running both directions after seeing the data inflates false-positive risk.

R pattern:

```
t.test(weight ~ group, data=sleep, alternative="two.sided")
```

Interpretation: Report the group estimates, confidence interval and p-value.

One-Sample t-Test

A one-sample t-test evaluates whether a sample mean differs from a reference value.

How it works	Decision rule	Common pitfall
t.test(x, mu=reference) compares the observed standardised difference with the t distribution.	Use when observations are independent and the distribution is reasonably compatible with the test.	A large sample can make a tiny, unimportant difference statistically significant.

R pattern:

```
meatmeal <- subset(chickwts, feed=="meatmeal")$weight
t.test(meatmeal, mu=160)
```

Interpretation: Pair the p-value with the estimated difference and interval.

ANOVA

One-way ANOVA tests whether at least one group mean differs among three or more independent groups.

How it works	Decision rule	Common pitfall
It compares between-group variation with within-group variation using an F statistic.	If significant, follow with planned contrasts or a multiple-comparison method to locate differences.	A significant ANOVA does not say every group differs from every other group.

R pattern:

```
model <- aov(weight ~ feed, data=chickwts)
summary(model)
TukeyHSD(model)
```

Interpretation: ANOVA answers an overall question; post-hoc analysis answers which groups differ.

Lab 11: Descriptive Statistics, Correlation and Regression

Goal: A validated summary, correlation result, fitted model and diagnostic evidence.

Scenario: A health analytics team studies height, weight and a numeric outcome.

Mapping: LO6 · K1 · K2 · A1 · A3 · A4 · A6 · Level: Advanced

Step-by-step

1. Import the health dataset from the workbook.

2. Describe valid sample size, centre and spread.

3. Plot height versus weight and calculate Pearson correlation.

4. Fit a linear model and interpret slope and R-squared.

5. Inspect residuals and record one limitation.

Open the folder labs/lab-11-descriptive-correlation-regression/ and run lab-11.R from that working directory.

Acceptance criteria

- Correlation interpretation includes direction and strength.
- Model formula and prediction target are explicit.
- Diagnostic evidence is included before a conclusion.

Evidence to retain: completed Excel Evidence sheet, generated output files, R console verification message and the final R script.

Lab 12: Hypothesis Testing and ANOVA Capstone

Goal: A t-test, ANOVA, post-hoc comparison and revised analysis decision.

Scenario: A health-policy team compares groups and proposes a defensible model update.

Mapping: LO6 · K5 · A1–A10 · Level: Advanced

Step-by-step

1. Write H0, H1 and alpha before running any test.

2. Run the specified two-group t-test and report estimates, interval and p-value.

3. Run one-way ANOVA across three or more groups.

4. If warranted, run TukeyHSD and identify supported pairwise differences.

5. Diagnose one unintended outcome and implement a justified script update.

Open the folder labs/lab-12-hypothesis-anova-capstone/ and run lab-12.R from that working directory.

Acceptance criteria

- Every conclusion uses reject/fail-to-reject language correctly.
- ANOVA conclusion does not overstate which groups differ.
- The updated script reruns successfully and the audit note explains the change.

Evidence to retain: completed Excel Evidence sheet, generated output files, R console verification message and the final R script.

Quick R Reference

Task	R pattern
Inspect object	str(x); class(x); summary(x)
Missing values	is.na(x); mean(x, na.rm=TRUE)
Filter rows	df[df$value > threshold & !is.na(df$value),]
Import Excel	readxl::read_excel('lab-workbook.xlsx', sheet='Input_Data')
Export CSV	write.csv(result, 'analysis_output.csv', row.names=FALSE)
Correlation	cor(x, y, use='complete.obs')
Regression	lm(outcome ~ input1 + input2, data=df)
t-test	t.test(value ~ group, data=df)
ANOVA	aov(value ~ group, data=df)

Assessment and Support

Written Assessment (SAQ) — 7 open-ended questions, K1–K7, 60 minutes, open book.

Case Study (CS) — 6 open-ended tasks, A1–A10, 80 minutes, open book.

Step	Action
1	Scan the TRAQOM/SSG QR code.
2	Complete assessment digital attendance.
3	Sit the WA then Case Study, open book.
4	Upload completed answer documents to the LMS.
5	Sign the Assessment Summary Record.

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